

REGIONAL BICYCLE PLAN

June 2010



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EASTGATE REGIONAL COUNCIL OF GOVERNMENTS

Serving Northeast Ohio since 1973

The Eastgate Regional Council of Governments is a multipurpose Regional Council of Governments for Ashtabula, Mahoning and Trumbull Counties, as established by Section 167.01 of the Ohio Revised Code. Eastgate is the agency designated or recognized to perform the following functions:

- Serve as the Metropolitan Planning Organization (MPO) in Mahoning and Trumbull Counties, with responsibility for the comprehensive, coordinated and continuous planning for highways, public transit and other transportation modes as defined in the Safe Accountable Flexible and Efficient Transportation Equity Act - A Legacy for Users (SAFETEA-LU).
- Perform continuous water quality planning functions in cooperation with Ohio and U.S. EPA.
- Provide planning to meet air quality requirements under SAFETEA-LU and the Clean Air Act Amendments of 1990.
- Administration of the Economic Development District Program.
- Administration of the Local Development District of the Appalachian Regional Commission.
- Administration of the State Capital Improvement Program for the District 6 Public Works Integrating Committee.
- Administration, in coordination with the Mahoning Valley Economic Development Corporation, of the Department of Defense Procurement Program.
- Administer the area clearinghouse function, which includes providing local government with the opportunity to review a wide variety of local or state applications for federal funds.
- Administration of the regional Rideshare Program for Ashtabula, Mahoning and Trumbull Counties.
- Conduct demographic, economic, and land use research.
- At General Policy Board direction, provide planning assistance to local governments that comprise the Eastgate planning area.
- Administration of the Clean Ohio Conservation & Revitalization Funds.

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Regional Bicycle Plan

1 | Introduction

The Eastgate Regional Council of Governments (Eastgate) is the official Metropolitan Planning Organization (MPO) for Mahoning and Trumbull Counties. Eastgate is responsible for a variety of federal, state, and local planning and project implementation programs. As the MPO and Areawide Water Quality Management Agency for Mahoning and Trumbull Counties, and the designated Economic Development District, Eastgate continues to maintain required certifications and planning documents to qualify the region for federal and state funding.

Other major areas of responsibility include: air quality planning and air advisory day programs, State Capital Improvement Program administration for the District 6 Public Works Integrating Committee, administration of the Department of Defense Procurement Program, Intergovernmental review, administration of the regional Rideshare program, administration of the Clean Ohio Conservation & Revitalization Funds and administration of the Local Development District of the Appalachian Regional Commission.

As the MPO, it is Eastgate's responsibility to maintain a Long Range Transportation Plan (LRTP) for its planning area; Mahoning and Trumbull Counties. The LRTP will provide for logical development within the region and emphasize the transportation system's effects within the area's physical, demographic, social and economic environment. The next LRTP will be finalized in 2013 and will outline a plan for the region through the year 2040.

The Regional Bicycle Plan will outline regional bicycle needs and priorities for the region. It will be revised as needed until incorporated into the 2040 LRTP. Regional plans are necessary to promote smart and cooperative regional development, but the Regional Bicycle Plan and the LRTP will not be binding static plans.

Eastgate respects the changing needs and desires of its local communities and will adapt its priorities and plans accordingly. It is important to note that the development, construction, and maintenance of a bicycle facility is largely dependent upon the sponsoring agency and not Eastgate. Eastgate's role is strictly related to planning and funding.

The current draft of the Regional Bicycle Plan outlines the short term goals related to completing the plan, and some long term goals for bicycle transportation in the region. It also outlines existing, planned and conceptual bikeways in the Eastgate area, identifies bicycle-vehicle accident locations, evaluates crash data, and summarizes bicycle use through census data. For communities interested in partnering with Eastgate, it also identifies funding options and outlines the project development process.

"The Eastgate Regional Council of Governments is the official Metropolitan Planning Organization for Mahoning and Trumbull Counties."

2|Goals

Eastgate plans to take specific steps in an effort to promote a safe multi-modal transportation network inclusive of bicyclists. Eastgate's goals include identifying a regional bicycle network, providing planning and funding to see the network to development, encouraging routine accommodation for bicyclists, identifying safety projects, promoting bicycling as a means of transportation, supporting educational outreach, and collecting data useful for bicycle planning.

Goals
Multi-modal Travel
Planning and Funding
Safety

Multi-modal Travel

Eastgate supports the creation of a regional bicycle and pedestrian network that connects every community, regional transit, major activity centers and central business districts. Eastgate is currently identifying existing conditions and deficiencies in regional multi-modal travel. Eastgate will encourage routine accommodation of bicycles and pedestrians in all transportation projects in an effort to create connections between all modes of transportation.

Planning and Funding

Eastgate will continue to provide planning and funding for the development of facilities that contribute to a bicycle friendly environment. Eastgate is committed to the completion of the Great Ohio Lake-to-River Greenway and supports other regional bikeway initiatives. The use of national design standards will be encouraged to facilitate seamless connections with other regions. Eastgate will also maintain an active discussion between the Federal Highway Administration (FHWA), the Ohio Department of Transportation (ODOT), the MPO and local communities regarding bicycle planning.

Safety

Eastgate will encourage local and statewide policies and projects that improve bicycle safety and will work with established bicycle groups to provide training sessions, promotional events, and educational outreach. Eastgate will continue to maintain and evaluate bicycle-vehicle crashes for patterns in location and circumstances and will identify locations and roadway geometrics needing safety upgrades for bicyclists.

3 | Timeline

Table 1: Timeline of activities that will support the development of the Regional Bicycle Plan.

Fiscal Year 2013		Sept																		
		Aug																		
Fiscal Year 2012		July																		
		June																		
	2012	May																		
		Apr																		
		Mar																		
		Feb																		
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	2011	Nov																		
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		Jan																		
		Dec																		
	2010	Nov																		
		Oct																		
		Sept																		
		Aug																		
		July																		

4|The Eastgate Regional Council of Governments

History and Structure

Over the years, Eastgate has progressed and advanced from a narrow planning agency based on highways only, to a broader multi-modal perspective. Eastgate's roots began with the Mahoning-Trumbull Counties Comprehensive Transportation and Development Study legislative committee in 1963. This group was composed of city, county, state and federal officials who developed a prospectus for transportation planning for the two-county area.

The Comprehensive Transportation and Development Study was formed in response to the Federal Highway Act of 1962. The Act formally recognized that transportation needs are regional, extending over physical boundaries or governmental units, and that due to this regional nature potential benefits could be derived from a comprehensive, coordinated and continuous planning process.

The 1962 Act directed the U.S. Department of Commerce, which at the time handled transportation matters, to assist the states in developing long-term highway development plans. These plans were to then be coordinated with other aspects of the overall transportation system. Further support for coordinated transportation planning came with the passage of the Urban Mass Transportation Act of 1964. This act made the availability of federal funds for highway and mass transit systems contingent upon the existence of comprehensive transportation plans within the areas to be served by the transit systems.

The State of Ohio, through the Ohio Department of Highways, in 1964 entered into an agreement with the Comprehensive Transportation and Development Study for preparation of the Comprehensive Transportation Plan and in 1965 local efforts were initiated utilizing state development procedural manuals. As of July 1, 1965, all Federal Aid Programs and projects proposed in metropolitan areas with a population over 50,000 had to be consistent with regional development plans, or approval would not be granted.

In 1973, Eastgate (initially known as the Eastgate Development and Transportation Agency) was formed as a council of governments under Section 167.01-08 of the Ohio Revised code. Eastgate was a voluntary organization formed through intergovernmental agreements enacted by its members whose purpose was to coordinate all local plans and recommendations in a cooperative effort for regional planning, programming and implementation. When Eastgate was formed the operations and functions of the Mahoning and Trumbull Counties Comprehensive transportation and Development Study were integrated within the organization structure of Eastgate.

“Eastgate is the regional voice of government, designated by the State of Ohio, with members including Mahoning County, Trumbull County, and all cities, villages, and townships in the counties.”

Today, Eastgate is more than just a transportation and development planning agency. Eastgate is the regional voice of government, designated by the State of Ohio, with members including Mahoning County, Trumbull County, and all cities, villages, and townships in the counties. Services are also provided to Ashtabula County through Appalachian Regional Commission, Ohio Rideshare and Economic Development Agency programs. Eastgate brings local governments together to create a unified voice in areas such as transportation, water and air quality, land use planning and local infrastructure projects.

“The General Policy Board is composed of elected officials representing the political jurisdictions of the planning area.”

The policy making board of Eastgate is the General Policy Board (GPB), which is composed of elected officials representing the political jurisdictions of the planning area. The duties of the GPB include recommending and coordinating local plans, policy statements and service programs for implementation by Eastgate, member agencies or contractors. Eastgate also uses three other planning committees to make recommendations and assist the Board in its decisions.

The Finance Committee is comprised of elected officials from the General Policy Board and is responsible for budget development and review as well as fiscal recommendations to the GPB.

The Technical Advisory Committee (TAC) consists of engineering and planning officials from the area. The TAC provides the General Policy Board with a broad base of technical expertise and assists the board with technical reviews and studies, as well as any in depth investigations requested by the Board, its standing committee or the Director.

The Citizens Advisory Board (CAB) is comprised of residents from the area and is responsible for considering methods and techniques of maximizing and insuring citizen representation and participation in Eastgate's planning activities.

In addition to these standing committees, the General Policy Board may also establish Ad Hoc Committees to address specific needs. The Board has created Ad Hoc Committees in the past to monitor the progress of special highway programs, assist with the preparation of air quality plans and assist with the development of coordinated transit programs.

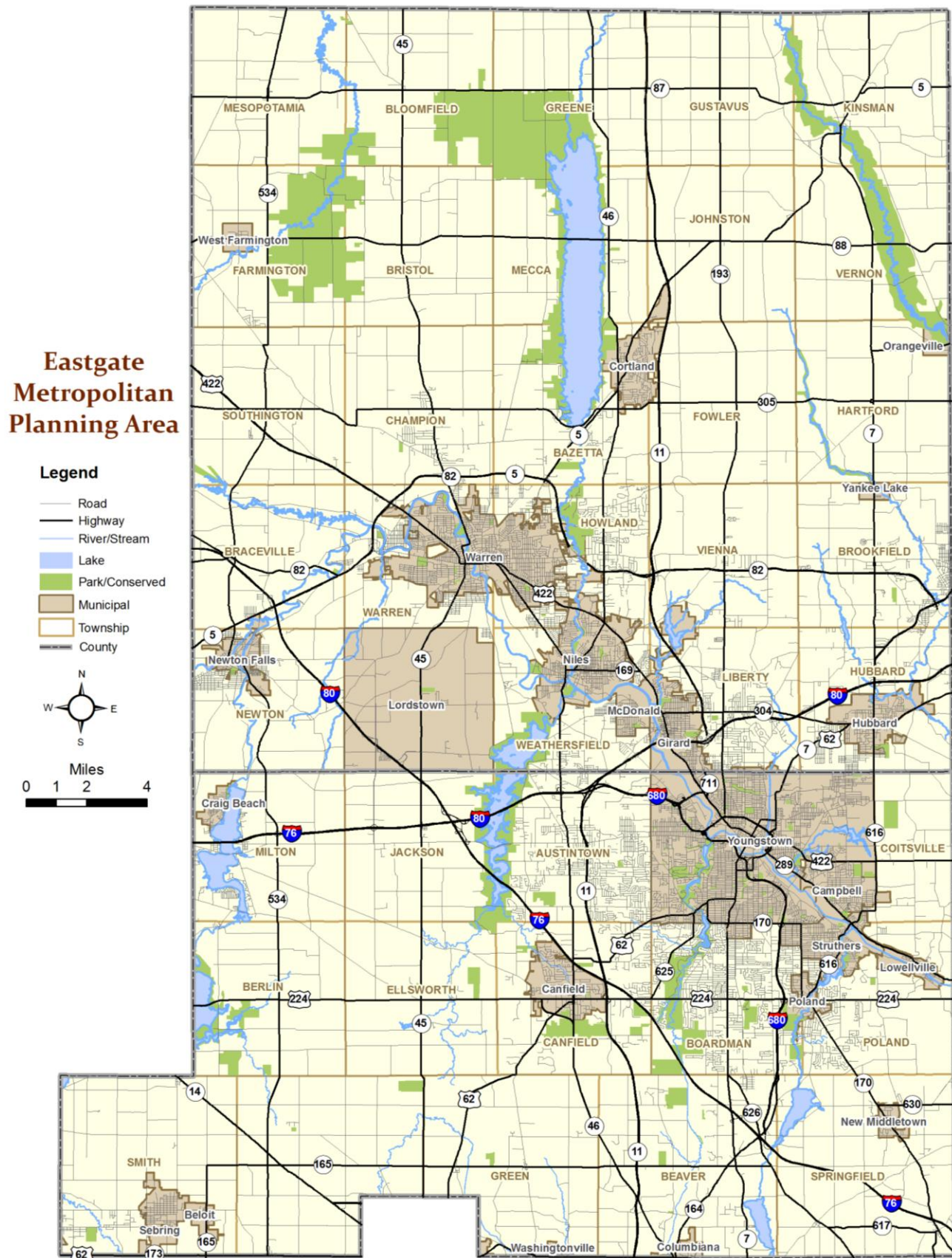
Planning Area

Eastgate provides services to Mahoning, Trumbull and Ashtabula Counties, all of which are located in Northeast Ohio. The majority of Eastgate's services consist of metropolitan planning activities for Mahoning and Trumbull Counties (Figure 1).

Mahoning and Trumbull Counties have both urban and rural communities, which makes the area suitable for both the advanced cyclist comfortable on busy urban streets and for the novice rider who prefers the wide shoulders and infrequent traffic often found in rural areas. The Youngstown-Warren metropolitan area is the primary urban focus of the counties and is composed of several cities clustered together.

The topography of the counties is relatively flat and favorable for bicycling; however Mahoning County does have more topographical variation than Trumbull County. The counties have a wide variety of land uses which offers the bicyclist the choice of riding roadways through scenic rural areas or through urban areas to commercial centers.

Figure 1: Eastgate Metropolitan Planning Area.



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Existing and Planned Bikeways

1|Existing and Planned Bikeways

The Eastgate planning area contains a number of bike paths, bike lanes and bike routes. Bikeways in the two-county area vary from small loop paths in local parks, to community-length trails, to cross-county bikeways with regional connections. Figure 2 outlines the existing and planned bikeways in Mahoning and Trumbull Counties.

The Great Ohio Lake-to-River Greenway

One of the most noted bikeways in the area is the Great Ohio Lake-to-River Greenway. The Greenway is a 10-foot wide asphalt shared-use path being built on former railroad lines and adjacent corridors. The portions of the Greenway constructed in Eastgate's planning area include the Western Reserve Greenway in Trumbull County and the Mill Creek MetroParks Bikeway in Mahoning County (Figure 3).

The Warren Greenway Phase I and the Niles Greenway are currently under construction. Several more sections of the Great Ohio Lake-to-River Greenway are programmed and will help to complete the regional bikeway. The sections currently programmed include the Warren Greenway Phase II and the Mill Creek MetroParks Bikeway Phase III.

"The Great Ohio Lake-to-River Greenway will be a 100 mile multi-modal transportation corridor which will connect Lake Erie to the Ohio River."

The Great Ohio Lake-to-River Greenway, when completed, will be a 100 mile multi-modal transportation corridor which will connect Lake Erie to the Ohio River (Figure 4). The Lake-to-River Greenway will begin on the shores of Lake Erie at Ashtabula Harbor, pass through Ashtabula, Trumbull, Mahoning and Columbiana Counties, and end at the banks of the Ohio River in East Liverpool.

The Greenway has been a regional effort, with different communities and organizations taking the lead on various sections of the bikeway. As a result, a majority of the Great Ohio Lake-to-River Greenway is either constructed or programmed for construction in the near future. The portion of the Greenway through the Eastgate area is nearly complete.

Figure 3: The Kirk Road Trailhead on the Mill Creek MetroParks Bikeway.



Figure 2: Existing and Planned Bikeways.

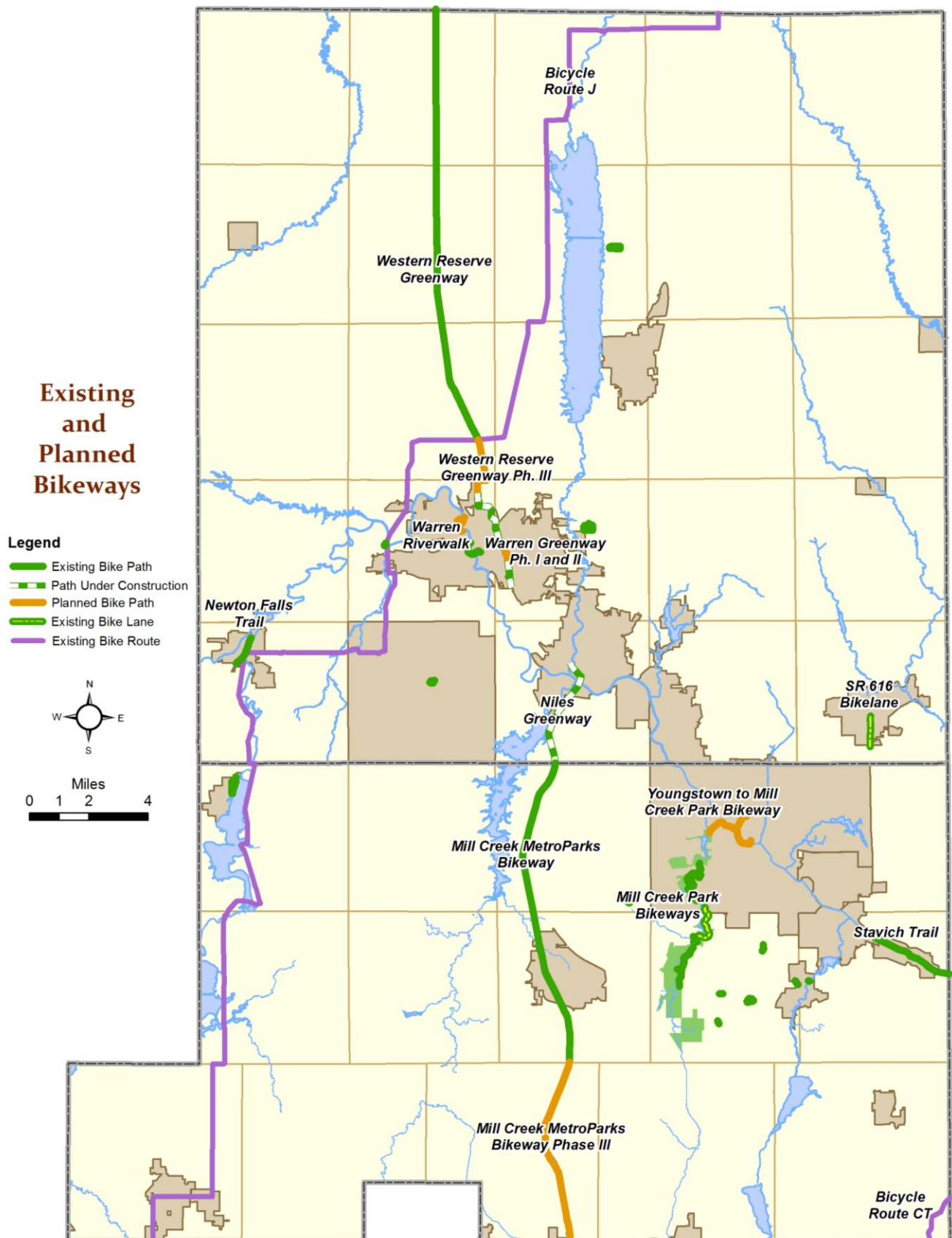
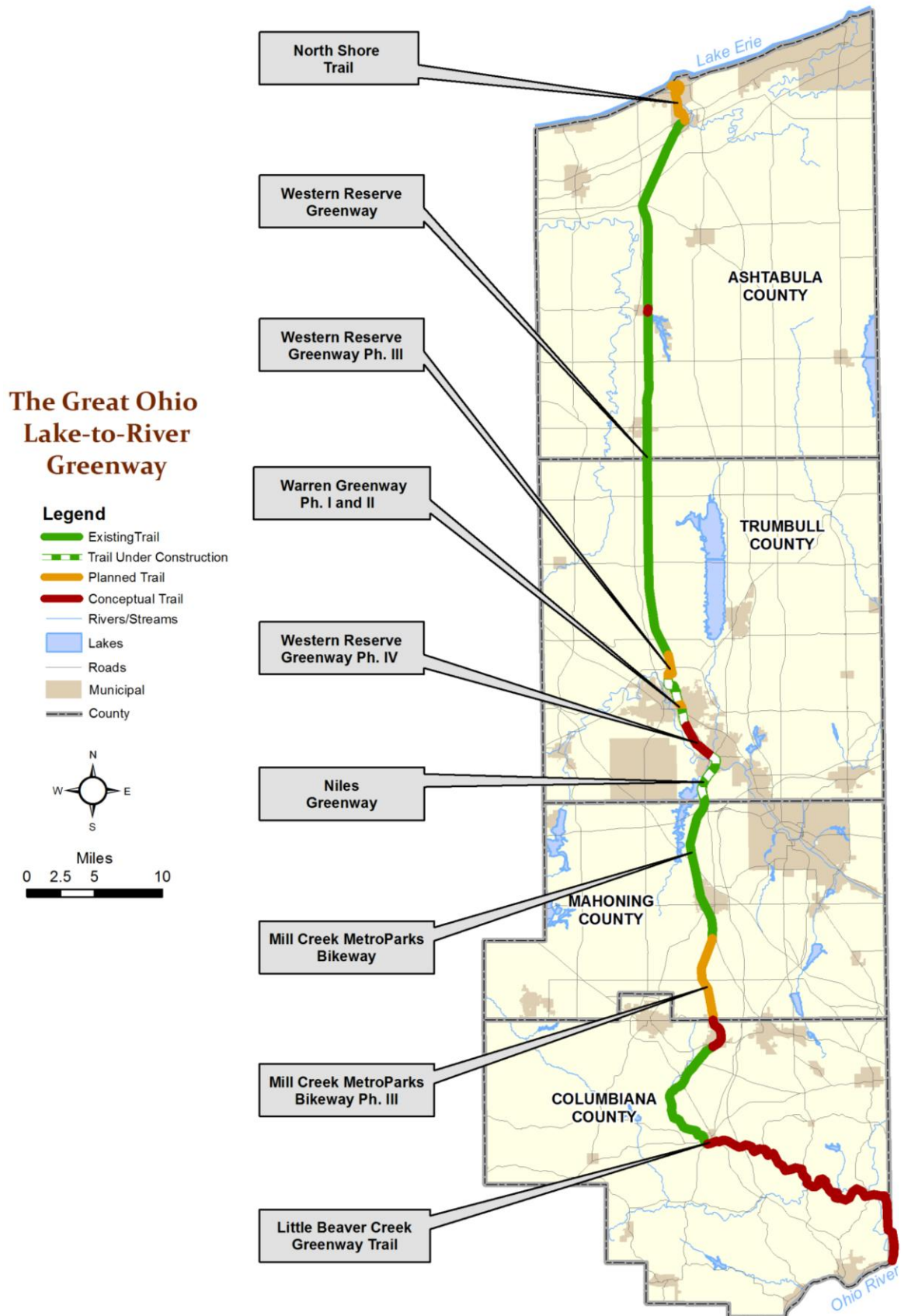


Figure 4: The Great Ohio Lake-to-River Greenway.



Constructed Greenway

Eastgate provided funding and planning support for several sections of the Great Ohio Lake-to-River Greenway which are now complete or are currently under construction. The Western Reserve Greenway and the Mill Creek MetroParks Bikeway have been completed. The Niles Greenway and the Warren Greenway are currently under construction.

The Western Reserve Greenway

The Western Reserve Greenway begins near West Avenue in the City of Ashtabula in Ashtabula County. It continues south into Trumbull County and terminates at Champion Street East, just north of the City of Warren. It is a total of 43 miles long, with 16 of those miles in Trumbull County.

The Mill Creek MetroParks Bikeway

The MetroParks Bikeway (Figure 5) is 11 miles long. It begins at the Mahoning-Trumbull County border in Austintown Township and continues south to Western Reserve Road in Canfield Township.

Figure 5: Mill Creek MetroParks Bikeway bridge over Mahoning Avenue.



The Niles Greenway (under construction)

The Niles Greenway will connect the City of Niles to the existing Mill Creek MetroParks Bikeway. It will begin near Robbins Avenue in downtown Niles and will tie into the MetroParks Bikeway at the Mahoning-Trumbull County line.

The Warren Greenway Phase I (under construction)

The completed Warren Greenway will run from North River Road south through the City of Warren to Burnett Street East. Phase I will complete most of the Greenway except for a short section downtown.

Planned Greenway

Eastgate has committed funds to upcoming Western Reserve Greenway Phase III, the Warren Greenway Phase II, and the Mill Creek MetroParks Bikeway Phase III.

The Western Reserve Greenway Phase III

Phase III will extend the existing Western Reserve Greenway south to the City of Warren. It will begin at Champion Street East, just north of Warren, and end at North River Road and the Warren City limits.

The Mill Creek MetroParks Bikeway Phase III

Phase III will extend the MetroParks Bikeway for 6.25 miles to the south, beginning at Western Reserve Road where the existing bikeway ends. Phase III will end at the Columbiana County line in Washingtonville.

The Warren Greenway Phase II

Phase II will complete the Warren Greenway through the central downtown area.

Conceptual Greenway

The last phase of the Greenway through the Eastgate area will be the Western Reserve Greenway Phase IV. Initial discussions on this project have begun, but official plans and funding sources have yet to be determined. Eastgate is committed to completing the Greenway and will provide funding and planning support to the Phase IV project sponsor.

Western Reserve Greenway Phase IV

Phase IV will connect the City of Warren at Burnett Street East to Robbins Avenue in the City of Niles.

The Stavich Trail

The Stavich Trail is a 12-mile long, 8-foot wide, paved bicycle and pedestrian trail running parallel to the Mahoning River and the rails of the former Youngstown-New Castle streetcar line (Figures 6-7). The trail originates at Broad Street in Struthers, Ohio, and continues through Lowellville and Poland Township in Mahoning County. From Poland Township, the Trail enters Pennsylvania in Hillsville, and passes through Union Township on its path to New Castle in Lawrence County, Pennsylvania. It ends at Washington Street, just outside of New Castle.

Figure 6: The Stavich Trail.



Figure 7: The Stavich Trail



The Newton Falls Trail

The Newton Falls Trail is a 3/4-mile long, 4-foot wide paved trail through downtown Newton Falls (Figure 8).

Figure 8: The Newton Falls Trail.



The Warren Riverwalk

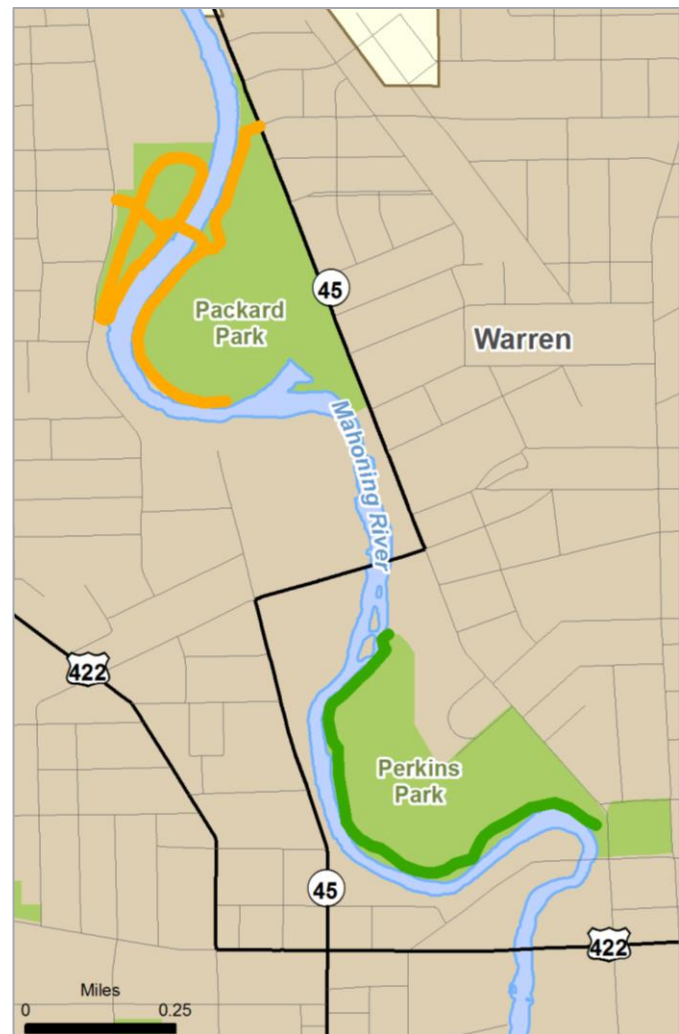
The Warren Riverwalk will be a 5-mile linear park along the Mahoning River in Warren with many bicycle and pedestrian components. Portions of the Riverwalk have already been completed and other sections are currently planned (Figures 9-10).

The Riverwalk will not only serve as a bicycle and pedestrian link through Warren, but when complete will connect Perkins and Packard Park, the Warren Community Amphitheater, the historic Kinsman House and possibly other bikeways, including the Great Ohio Lake-to-River Greenway.

Figure 9: The Warren Riverwalk, Image courtesy of the City of Warren.



Figure 10: The Warren Riverwalk



The Hubbard Bike Lane

The bike lane along SR 616 in Hubbard is a 2-mile long shoulder designated bike lane (Figures 11-12).

Figure 11: The Hubbard Bike Lane



Figure 12: The Hubbard Bike Lane



The Mill Creek Park Bikeways

Mill Creek Park offers many bicycle facilities within its 2,882 scenic acres. The park has 20 miles of low speed roadways which attract many bicyclists. It also has many facilities designated for bicyclists, including bike paths closed to motorized traffic, bike lanes and pedestrian bridges (Figure 13). The bikeways existing in Mill Creek Park are:

Youngstown to Mill Creek Bikeway

The Mill Creek MetroParks is currently planning a bikeway project which would connect Youngstown State University, the Covelli Center and Mill Creek Park.

East Cohasset Hike/Bike Trail

The East Cohasset Hike/Bike Trail is a 1.5-mile asphalt hike/bike path that is closed to motorized traffic. It provides a scenic route above Lake Cohasset.

East Newport Hike/Bike Trail

The East Newport Hike/Bike Trail is a 1.75-mile long, two way hike/bike lane alongside a one-way road. The trail connects to West Newport Drive on the opposite side of Lake Newport via a pedestrian-only bridge across Mill Creek.

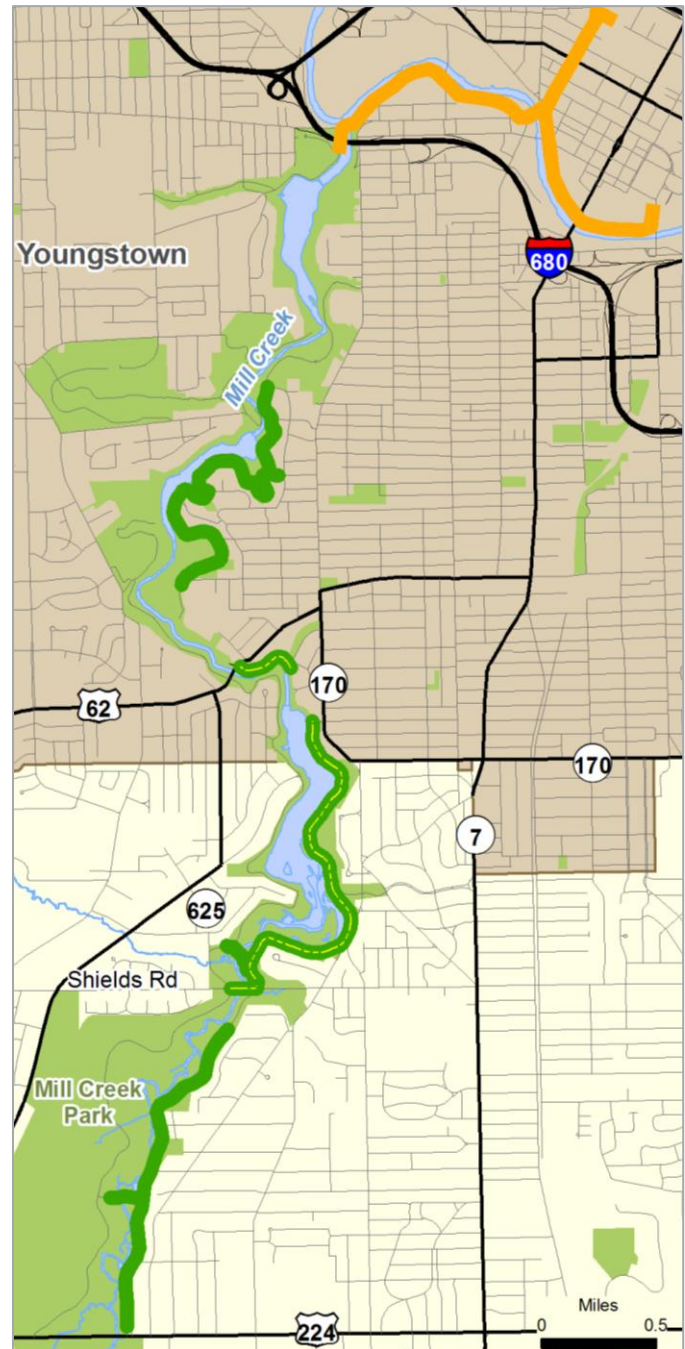
Shields Road Bridge

Shields Road crosses through Mill Creek Park, with the East Golf Hike/Bike Trail to the south and the East Newport Hike/Bike Trail to the north. Bicycle safety was a concern at this intersection and a bike lane was installed on both sides on the Shields Road bridge over Mill Creek. Bicyclists now have a safer way to access both hike/bike trails.

East Golf Hike/Bike Trail

The 1.5-mile asphalt East Golf Hike/Bike Trail runs from Route 224 to Shields Road. Parking lots are located at each end of the trail and a pedestrian bridge across Mill Creek connects the trail with Golf Course amenities and additional parking.

Figure 13: The Mill Creek Park Bikeways



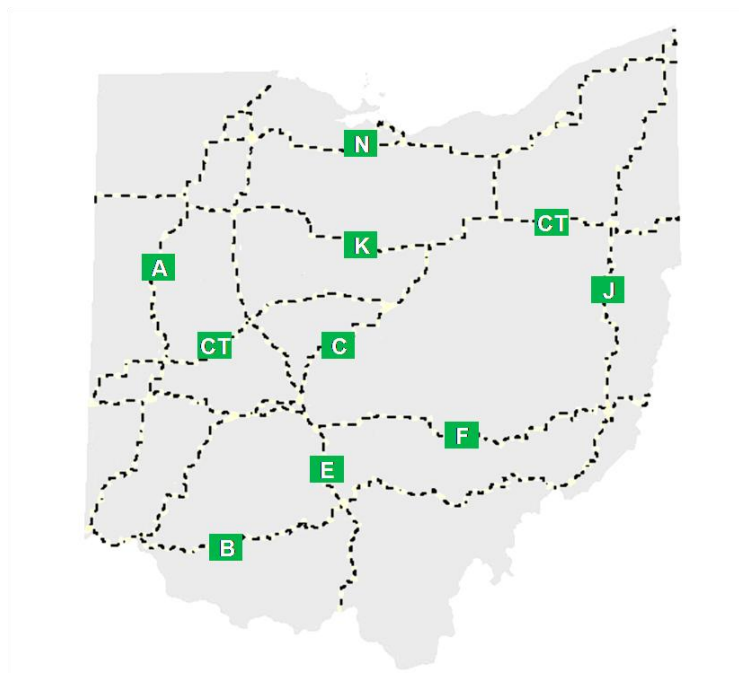
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2|Bicycle Routes

Ohio has 9 cross-state bicycle routes which incorporate low traffic volume roads and shared-use paths (Figure 14). Two of the cross-state bicycle routes, Route J and Route CT (Cardinal Trail), pass through the Eastgate planning area (Figure 15).

Bicycle routes utilizing low volume roadways are good alternatives to high cost shared-use paths. Bicycle routes require no additional right-of-way, striping, or infrastructure. Signage is the only item required, serving the dual purpose of providing way finding and reminding drivers to be alert for cyclists.

Figure 14: Ohio Bicycle Routes



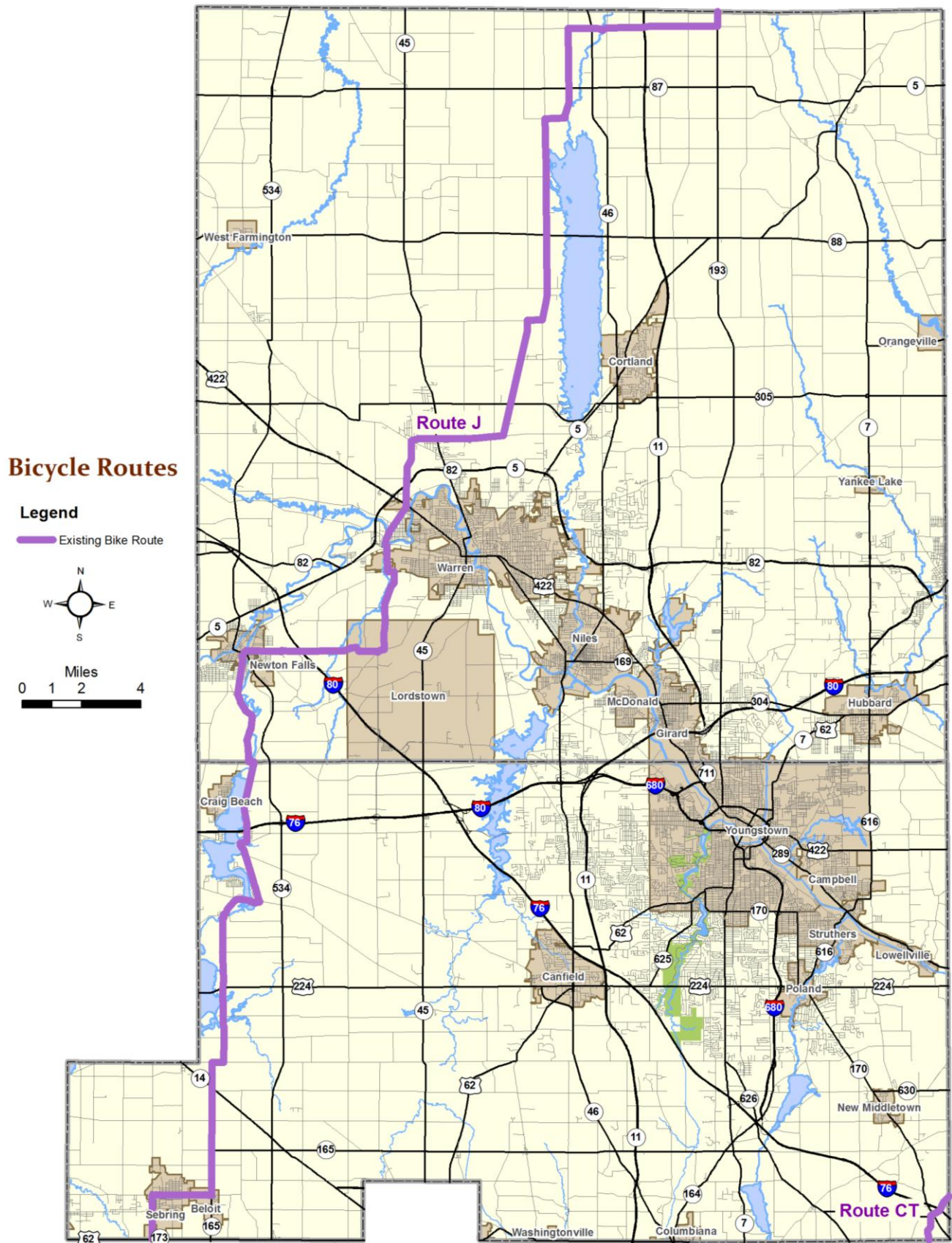
Bicycle Route J

Bicycle Route J connects Marietta, Ohio along the Ohio River to Conneaut, Ohio along Lake Erie. The Trumbull County portion of Route J has signs to identify the route. The bicycle route continues through western Mahoning County.

Bicycle Route CT

Bicycle Route CT runs from Richmond, Indiana to Petersburg, Ohio. The Cardinal Trail only passes through a small portion of Mahoning County at the southeast corner.

Figure 15: Bicycle Routes.



The Conceptual Network

1|Introduction

The vision of expanding the existing bicycle network within our region and making connections with neighboring counties and states has appeal. Regional bikeways would serve as an alternative transportation option and at the same time provide recreational opportunities. Many conceptual bikeway ideas currently exist in Mahoning and Trumbull County (Figure 16).

The conceptual bikeways do not represent future plans. The amount of discussion on and feasibility regarding each conceptual bikeway varies greatly. Some of the conceptual bikeways are very feasible and are simply waiting for leadership and funding. Others are simply ideas that represent a vision of future regional connections. Locations of conceptual bikeways represent locations where regional connections could be made, where existing bikeways could be expanded, where spurs from existing or planned bikeways would be ideal and where abandoned or active railways exist.

2|Current Concepts

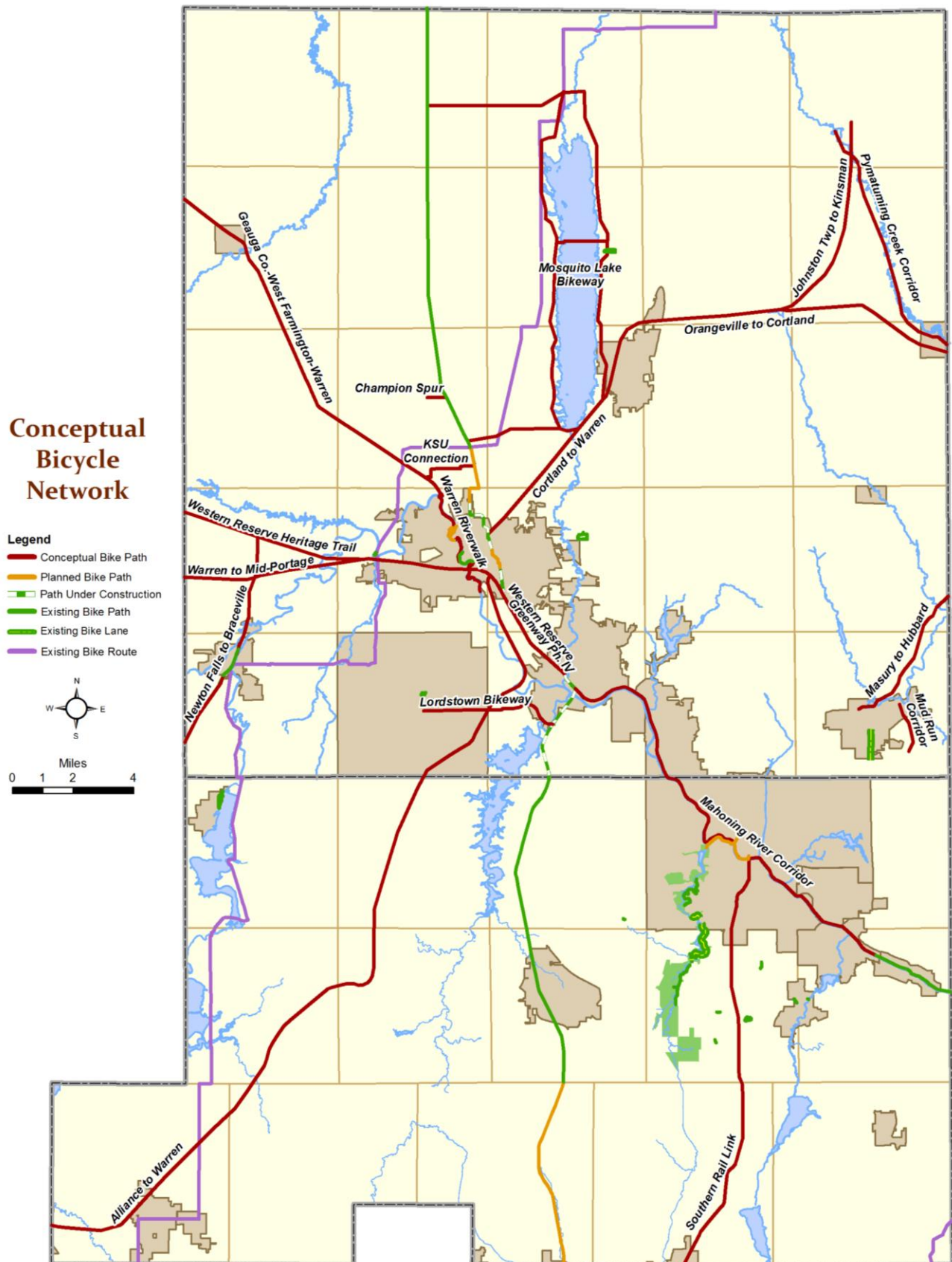
Some of the conceptual bikeways that represent the possibility of regional connections include the Western Reserve Heritage Trail and the Mahoning River Corridor Trail. These trails would connect the cities of Cleveland and Pittsburgh via the Mahoning Valley. The regional corridor would utilize and connect several trails including the Portage County MetroParks' Headwaters Trail, the Great Ohio Lake to River Greenway and the Stavich Trail. Much of the proposed trail would ideally follow abandoned rail rights of way.

Existing bikeways provide obvious locations for conceptual extensions. A conceptual extension exists for the Newton Falls Trail which would connect it to other conceptual bikeways. The Warren Riverwalk also has conceptual extensions, both north and south, which would connect the Riverwalk area to the Great Ohio Lake-to-River Greenway and other conceptual bikeways.

There are several conceptual spurs off of the Great Ohio Lake-to-River Greenway which would provide local connections to this regional bikeway. Conceptual ideas include connections to schools (such as Champion High School and Kent State Trumbull), to local communities (Lordstown), and to local parks and recreation centers (Mosquito Lake).

Railways are often targeted for bikeways because railways are well graded, have right-of-way that is an ideal size for a bikeway, are long corridors, and usually have culverts, bridges, and base materials still in place. Many of the conceptual bikeways locations are based on railroad right-of-way. Some railways are still active but have conceptual bikeways proposed for their location should they ever go out of service. The conceptual Southern Rail Link connecting the Mahoning River and Columbiana County is one example of a conceptual bikeway utilizing a currently active railway.

Figure 16: Conceptual Bicycle Network.



Complete Streets

1|Introduction

Many positive changes have occurred since the last Regional Bicycle Plan was published by Eastgate in 1995. Not only have there been physical improvements, such as the completion of two major portions of the Great Ohio Lake to River Greenway, but there has been a national shift toward multi-modal transportation options and environmentally friendly transportation.

Bicycling is no longer seen as solely for recreation, partially due to a policy statement made by the U.S. Department of Transportation (USDOT). The policy statement and guide, *Accommodating Bicycle and Pedestrian Travel: A Recommended Approach* (USDOT, 2008), states that "bicycling and walking facilities will be incorporated into all transportation projects unless exceptional circumstances exist."

The policy statement was initiated and maintained in response to language in the latest federal transportation bills; the 2005 Safe, Accountable, Flexible, Efficient Transportation Equity Act – A Legacy for Users and the 1998 Transportation Equity Act for the 21st Century. Both bills indicate that safe accommodation of non-motorized users shall be considered during the planning, development and construction of all Federal-aid transportation projects and programs.

Recent policy changes by the USDOT and statements made by the Secretary of Transportation have declared that bicycling is a priority and equal with that of other transportation modes. The next federal surface transportation reauthorization bill is expected to reflect that sentiment and provide more opportunities for developing bicycle facilities.

In addition to the federal government, state and local agencies, planners, and engineers are also beginning to shift to a multi-modal mindset. States and communities are also beginning to adopt complete street policies. Complete street policies ensure that planners and engineers design for all users including bicyclists, public transportation vehicles and riders, and pedestrians of all ages and abilities (Figure 17).

ODOT is currently developing a complete street policy which will apply to transportation projects funded through Eastgate. Some modifications to ODOT's Transportation Engineering Manual have already taken effect. One recent change is that all traffic signal projects utilizing state or federal funds are now required to provide bicycle detection.

"All traffic signal projects utilizing state or federal funds are now required to provide bicycle detection."

Figure 17: Example of a complete street. Image courtesy of the National Complete Streets Coalition.



2|Evaluation

The evaluation of existing conditions and the development of priorities is one of the first steps in moving toward complete streets. A roadway inventory report was previously developed by Eastgate in 1997 to identify and prioritize conditions along roadways and document their potential for use as bicycle routes. The inventory report was developed in cooperation with area bicyclists who recommended that specific routes be available to provide service between local residential neighborhoods and public or commercial attractions.

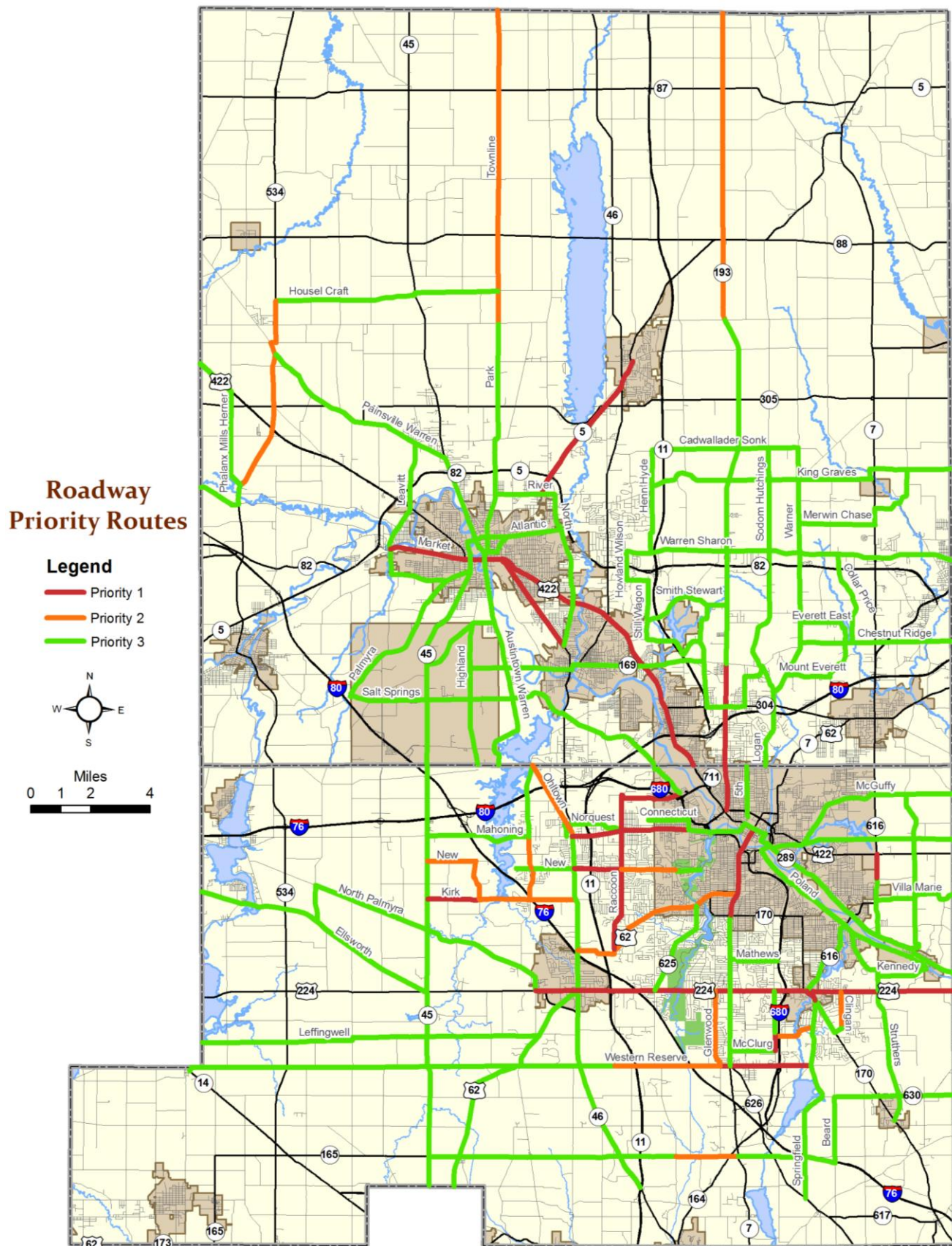
Eastgate analyzed and evaluated their recommendations while taking into consideration traffic volumes, posted speeds, observed speeds, width of outside lanes and the presence or absence of connectivity at activity centers. Roads were assigned a priority rating as follows:

Table 2: Priority ratings.

Priority 1	Route segments that will require major rehabilitation to be conducive to bicycle travel.
Priority 2	Route segments that are somewhat conducive to bicycle travel, but are in need of minor rehabilitation.
Priority 3	Route segments that are relatively better for bicycle travel, but planning activities and rehabilitation may still be warranted.

The priority ratings and resulting map (Figure 18) were updated to reflect major roadway improvements which have occurred since the original inventory was documented. However, ratings will receive a full revision to accurately reflect current circumstances and needs.

Figure 18: Roadway Priority Routes.



The new bicycle inventory will also show level-of-service (LOS), or how favorable conditions are for cycling on a particular roadway. Eastgate will review roadway segments, using methodology developed by other regional agencies, resulting in a detailed assessment of current roadway conditions.

The bicycle LOS map will serve several purposes:

1. It will show where inadequate conditions or broken connections exist.
2. It will serve as resource when developing and prioritizing a regional network plan.
3. It will serve as a route planning tool for cyclists.

The information in the inventory report will be transferred into a usable two-county bicycle map which will categorize roadways as appropriate for either the experienced, average or novice bicyclist. Eastgate's former method of categorizing cyclists and roadways will be reviewed and modified in the process. Bicyclists and roadways were formerly categorized as experienced, intermediate and novice using the following criteria:

Table 3: Cycling categories.

Experienced	Bicyclist is knowledgeable in bicycling and comfortable riding in traffic. A roadway rated as suitable for the experienced rider has one of the following conditions: 1. Fair or good pavement surface, an outer lane with of at least 12 feet and speeds not in excess of 45 mph; or 2. A paved clean shoulder at least 3-foot wide.
Average	Bicyclist rides in traffic but prefers to ride on the shoulder away from traffic. For a route to be considered compatible with these riders the roadway requires one of the following conditions: 1. A good pavement surface, an outer lane width of at least 14 feet (or 12 feet when more than one lane exists) and speeds not in excess of 35 mph; or 2. A 3-foot shoulder and speeds of 45 mph or less.
Novice	Bicyclist is uncomfortable riding in traffic and requires a right-of-way outside the flow of traffic. A roadway rated as suitable for the experienced rider has the following conditions: 1. A continuous paved shoulder in good condition that is at least 3-foot wide and traffic speeds less than 40 mph.

The LOS designations resulting from Eastgate's analysis will be modified by a group of experienced cyclists. Their recommendations will outweigh LOS designations derived from statistical means. Local cyclists have been informed of this project and are eager to participate.

3|Priorities

Bikeway and complete street priorities will be identified as part of an upcoming Modal Transportation Needs Assessment. The Needs Assessment will be based on current network inventories and conditions for the two-county area which will be analyzed and prepared over the next several work programs. Eastgate has already begun documenting and evaluating all multi-modal facilities and connections. Viewing all modes of transportation together will show exactly where connections are lacking and help determine key locations for investment.

Project recommendations will be developed in an effort to address problems and/or deficiencies identified as a result of the Modal Transportation Needs Assessment. The supporting framework for complete street planning is already in place at Eastgate through many current programs and groups. Eastgate has been partnering in Sustainable Communities, Complete Streets and Transit Connectivity initiatives already and will work within these current programs to identify and carry out projects.

Eastgate's Environmental Planning Advisory Committee (EPAC) will be a key group in identifying and initiating complete street projects. Bicycle specific projects and plans will also be reviewed through outreach to local bicycling groups such as the Outspoken Wheelman, the Great Ohio Lake-to-River Greenway Coalition, and Friends of the Western Reserve Greenway.

Eastgate is documenting all comments received from the public, cyclists, and local planners and engineers regarding recommended priorities. Formal meetings will take place as part of the LRTP process and provide an opportunity for public input regarding complete street and bicycle plans. Public outreach for specific projects will also be held the appropriate stage in project development.

Safety

1|Introduction

Data from the United States 2000 Census indicates that bicycles serve as a method of transportation to work for a small number of commuters in Mahoning and Trumbull County (Table 4). The census data provides insight regarding a portion of bicycle use, but does not reflect all of the bicycle use in the two-county area. The census data is limiting because it only identifies people's mode of transportation to work, outside of the home, for individuals over the age of 16.

Table 4: Mode of transportation to work for individuals over 16 years of age employed outside of the home (U.S. Census 2000).

Mode	U.S.A	Ohio	Mahoning/Trumbull Percent	People
Drove alone	78.2%	85.1%	88.6%	179,379
Carpooled	12.6%	9.6%	8.6%	17,425
Public transportation	4.9%	2.1%	0.6%	1,198
Motorcycle	0.1%	0.0%	0.1%	105
Bicycle	0.4%	0.2%	0.1%	138
Walked	3.0%	2.4%	1.5%	2,976
Other means	0.7%	0.5%	0.6%	1,236
Total	100.0%	100.0%	100.0%	202,457

The FHWA, as part of the 1994 National Bicycling and Walking Study, adopted the goal of increasing non-motorized transportation to at least 15 percent of all trips. Figures are not available which reflect all travel for all people in the two-county area, but it can be observed that bicycles are a common feature on our roadways at all times of the day and during all seasons of the year.

A goal of the FHWA, resulting from the 1994 National Bicycling and Walking Study, is to reduce the number of non-motorized users killed or injured in traffic crashes. The FHWA has requested that Eastgate review crash data to determine what safety issues, if any, exist in the Eastgate area.

According to the FHWA's Review of Ohio Bicycle and Pedestrian Planning and Safety Efforts (2009) common causes of bicycle accidents are bicyclists failing to stop at intersections or crossing into traffic, and vehicles striking bicyclists from behind.

"Common causes of bicycle accidents are bicyclists failing to stop at intersections..."

- Federal Highway Administration, 2009

Though accidents do happen, street riding is very safe. When a bicycle is in the street and follows the rules of the road, it is predictable part of traffic. Novice riders apprehensive about street riding may adopt dangerous practices that they think are safer, such as sidewalk riding, riding against the flow of traffic, or cutting through parking lots. These practices make the cyclist unpredictable and unexpected, leading to accidents.

A combination of education and bicycle facility improvement may help to alleviate the perceived fear of riding bicycles, increase driver awareness of bicycles, and correct any actual danger created by roadways that are not bicycle friendly. Eastgate will work with established bicycle groups to provide training sessions, promotional events, and educational outreach. Outreach should concentrate on riding according to rules of the road and educating motorists about a bicycle's legal right to the road.

Many changes in law, policy, and infrastructure that could make bicycling a safer and more enjoyable mode of transportation are not within Eastgate's power to administer. However, Eastgate will take advantage of the benefits derived by jointly assisting in project development and financial participation between the FHWA, ODOT, County Engineers, ODPS, and all of our local communities within our two county planning area to bring about as many positive changes as possible.

In the USDOT's 2002 National Survey of Pedestrian and Bicyclist Attitudes and Behaviors, 46.9 percent of participants reported that their community needed to make changes to improve bicycle safety. Suggestions that were reported included:

- Providing bicycle facilities: bicycle trails, paths, lanes, racks, traffic signals, lighting, or crosswalks (73.0%)
- Improving existing bicycle facilities (7.8%)
- Changing existing laws governing bicycles (7.3%)
- Initiating bicycle safety education (6.7%)
- Making areas for bicycling safer (6.0%)
- Enforcing laws governing bicycling (3.6%)
- Other suggestions (7.2%)

In an effort to create a safe cycling environment, Eastgate will continue to collect and monitor crash and commute data, will pursue funding through the ODOT Highway Safety Program, and will advise project sponsors to consider the safety of bicyclists while planning projects.

2|Crash Statistics

Eastgate obtains crash data each year from the Ohio Department of Public Safety for Mahoning and Trumbull Counties. Three years of crash data were reviewed for patterns in bicycle-related crashes. A three-year review of crash data is a standard timeframe used by ODOT when considering funding a safety project. A preliminary review of older crash data did not show significant differences in crash statistics.

Eastgate and its local communities strive to make bicycling a viable and safe means of transportation in an effort to reduce bicycle accidents, injuries and fatalities. Currently, bicycle-vehicle accidents in Mahoning and Trumbull County are around 50 per year (Table 5). Fortunately, most accidents do not result in a serious injury (Table 6).

On a regional level, safety is not a serious issue. However, a more detailed investigation reviewing patterns in crash locations and type will be conducted. Locations having a higher incidence of crashes will have original crash reports reviewed to determine the reason for crashes at that particular location.

Table 5: Total bicycle-vehicle crashes per year; Mahoning and Trumbull Counties.

Bicycle-Vehicle Accidents Per Year	Total
2005	52
2006	43
2007	49
2008	57
Total	201

Table 6: The injuries sustained by bicyclist involved in a crash with a vehicle; Mahoning and Trumbull Counties, 2006-2008.

Injury Severity Sustained by Bicyclist	Total	Percent
Fatal	1	0.7%
Incapacitating	22	14.8%
Non-incapacitating	63	42.3%
Possible	33	22.1%
No injury	16	10.7%
Unknown	14	9.4%
Total	149	100.0%

Analysis of crash data from 2006 through 2008 shows that most bicycle-vehicle accidents occur at intersections (Table 7) and that the bicyclist is most often cited as causing the crash (Table 8). Bicyclists cause accidents by failing to stop at intersections and improperly crossing into traffic. Educating the public about safe bicycling practices and bicycle laws may help to reduce the number of accidents.

Table 7: The location where a bicycle-vehicle crash occurred; Mahoning and Trumbull Counties 2006-2008.

Crash Location Type	Total	Percent
Intersection	86	57.7%
T-Intersection	45	30.2%
Four Way	40	26.8%
Y-Intersection	1	0.7%
Not and Intersection	44	29.5%
Driveway	17	11.4%
On-Ramp	2	1.4%
Total	149	100.0%

Table 8: The vehicle at fault and the error which contributed to the crash; Mahoning and Trumbull Counties, 2006-2008.

Vehicle at Fault and Contributing Circumstances	Total	Percent
Bicycle	95	63.7%
Unknown	24	16.1%
Improper Crossing	15	10.1%
Failure To Yield Right Of Way	13	8.7%
Darting	12	8.0%
Wrong Side Of Road	9	6.0%
Failure To Obey Traffic Signal/Sign	8	5.4%
Other	5	3.4%
None	5	3.4%
Inattentive	2	1.3%
Illegally In Roadway	1	0.7%
Not Visible	1	0.7%
Motorist at Fault	29	19.5%
Failure To Yield	11	7.4%
Followed Too Closely	6	4.0%
Unknown	5	3.4%
Inattention	3	2.0%
Ran Signal/Sign	2	1.3%
Improper Turn	1	0.7%
Reckless	1	0.7%
No Fault	25	16.8%
Total Crashes	149	100.0%

Most crashes occur during favorable conditions when more cyclists are on the road. This includes during fair weather months, during the day, and in clear weather (Tables 9-11). Over 80 percent of crashes happen during daylight hours (Table 12). More specifically, 60 percent of crashes occur between the hours of 2:00 and 7:00 PM. This timeframe corresponds with peak vehicle traffic. Though time of day is an indicator of crash likelihood, day of week is not. There is not a significant difference between weekday or weekend days (Table 13).

Table 9: Crashes by month, Mahoning and Trumbull Counties, 2006-2008.

Month when Crash Occurred	Total	Percent
January	5	3.4%
February	2	1.3%
March	9	6.0%
April	7	4.7%
May	14	9.4%
June	31	20.8%
July	19	12.8%
August	27	18.1%
September	16	10.7%
October	12	8.1%
November	3	2.0%
December	4	2.7%
Total	149	100%

Table 10: Crashes by light condition, Mahoning and Trumbull Counties, 2006-2008.

Light Condition when Crash Occurred	Total	Percent
Daylight	120	80.5%
Dark	19	12.8%
Dusk	6	4.0%
Dawn	0	0%
Glare	1	0.7%
Unknown	3	2.0%
Total	149	100%

Table 11: Crashes by weather condition, Mahoning and Trumbull Counties, 2006-2008.

Weather Condition when Crash Occurred	Total	Percent
Clear	101	67.8%
Rain	33	22.1%
Cloudy	8	5.4%
Snow	3	2.0%
Foggy	0	0%
Sleet/Freezing Rain	0	0%
Windy	0	0%
Unknown	4	2.7%
Total	149	100%

Table 12: Crashes by hour of day, Mahoning and Trumbull Counties, 2006-2008.

Time of Day when Crash Occurred	Total	Percent
12:00-1:00 AM	6	4.0%
1:00-2:00 AM	0	0%
2:00-3:00 AM	0	0%
3:00-4:00 AM	0	0%
4:00-5:00 AM	0	0%
5:00-6:00 AM	0	0%
6:00-7:00 AM	1	0.7%
7:00-8:00 AM	1	0.7%
8:00-9:00 AM	3	2.0%
9:00-10:00 AM	4	2.7%
10:00-11:00 AM	1	0.7%
11:00-12:00 AM	4	2.7%
12:00-1:00 PM	8	5.4%
1:00-2:00 PM	7	4.7%
2:00-3:00 PM	18	12.1%
3:00-4:00 PM	21	14.1%
4:00-5:00 PM	18	12.1%
5:00-6:00 PM	15	10.1%
6:00-7:00 PM	18	12.1%
7:00-8:00 PM	9	6.0%
8:00-9:00 PM	8	5.4%
9:00-10:00 PM	2	1.3%
10:00-11:00 PM	4	2.7%
11:00-12:00 PM	1	0.7%
Total	149	100%

Table 13: Crashes by day of week, Mahoning and Trumbull Counties, 2006-2008.

Day of Week when Crash Occurred	Total	Percent
Sunday	17	11.4%
Monday	23	15.4%
Tuesday	19	12.8%
Wednesday	19	12.8%
Thursday	20	13.4%
Friday	28	18.8%
Saturday	23	15.4%
Total	149	100%

3|Crash Locations

Bicycle-vehicle crashes cluster in urban areas, where there are more people, more traffic, and more destinations. Accidents happen on all types of streets including residential streets, state routes, and US routes (Figures 19 to 21).

Accident patterns may indicate where safety upgrades, signage, or sharrows may be needed. Accidents can be seen along major travel corridors such as SR 7, Mahoning Avenue, South Avenue and US 422. Corridors such as these are actively being used by bicycles. Ensuring that bicycles are considered in future upgrades and that the routes remain available during construction should be a priority.

Other safety issues and priority locations have been mentioned by the public, cyclists, and local planners and engineers. Eastgate will be considering all comments received and will be soliciting public input in preparation for the 2040 LRTP.

Figure 19: Warren-Niles Area Bicycle-Vehicle Accidents.

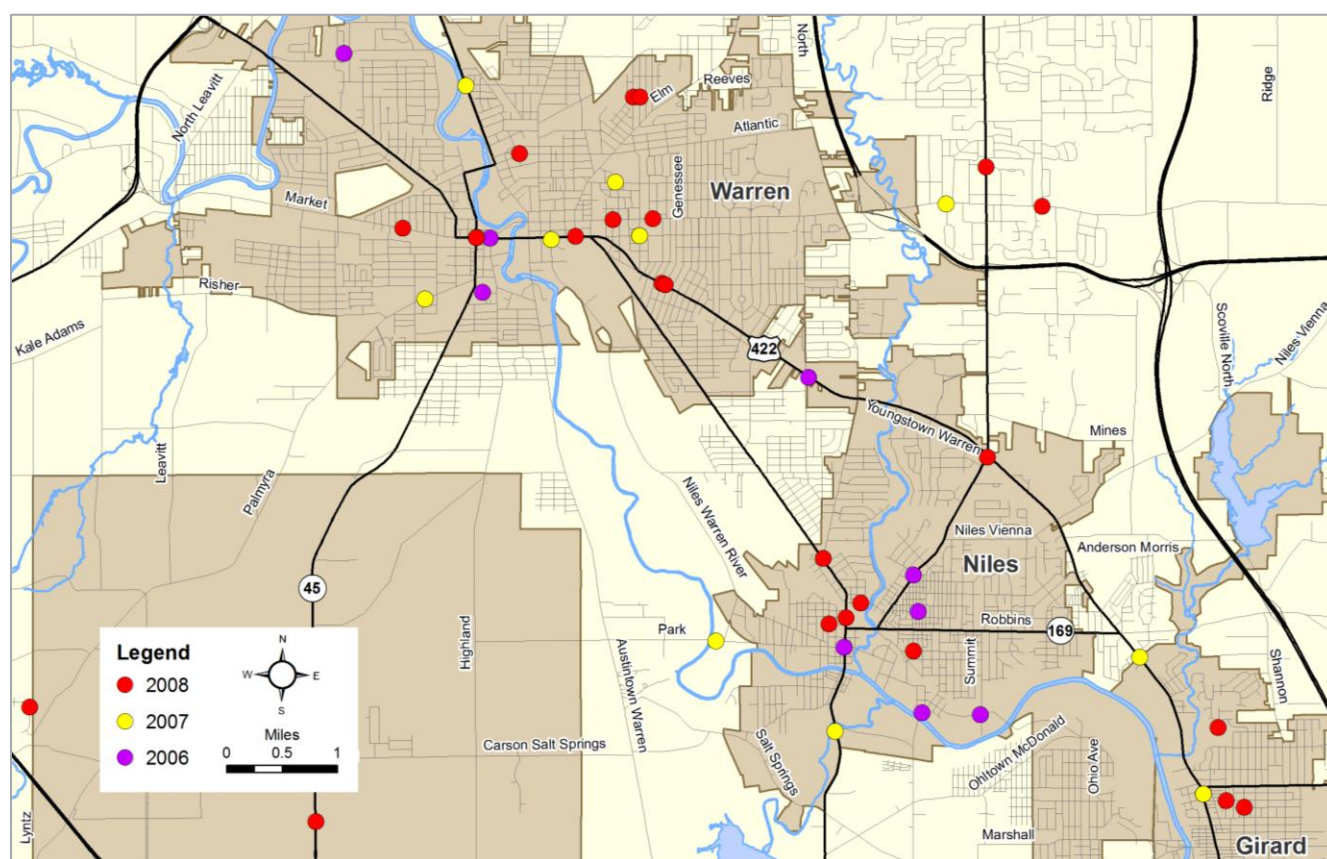


Figure 20: Youngstown Area Bicycle-Vehicle Accidents.

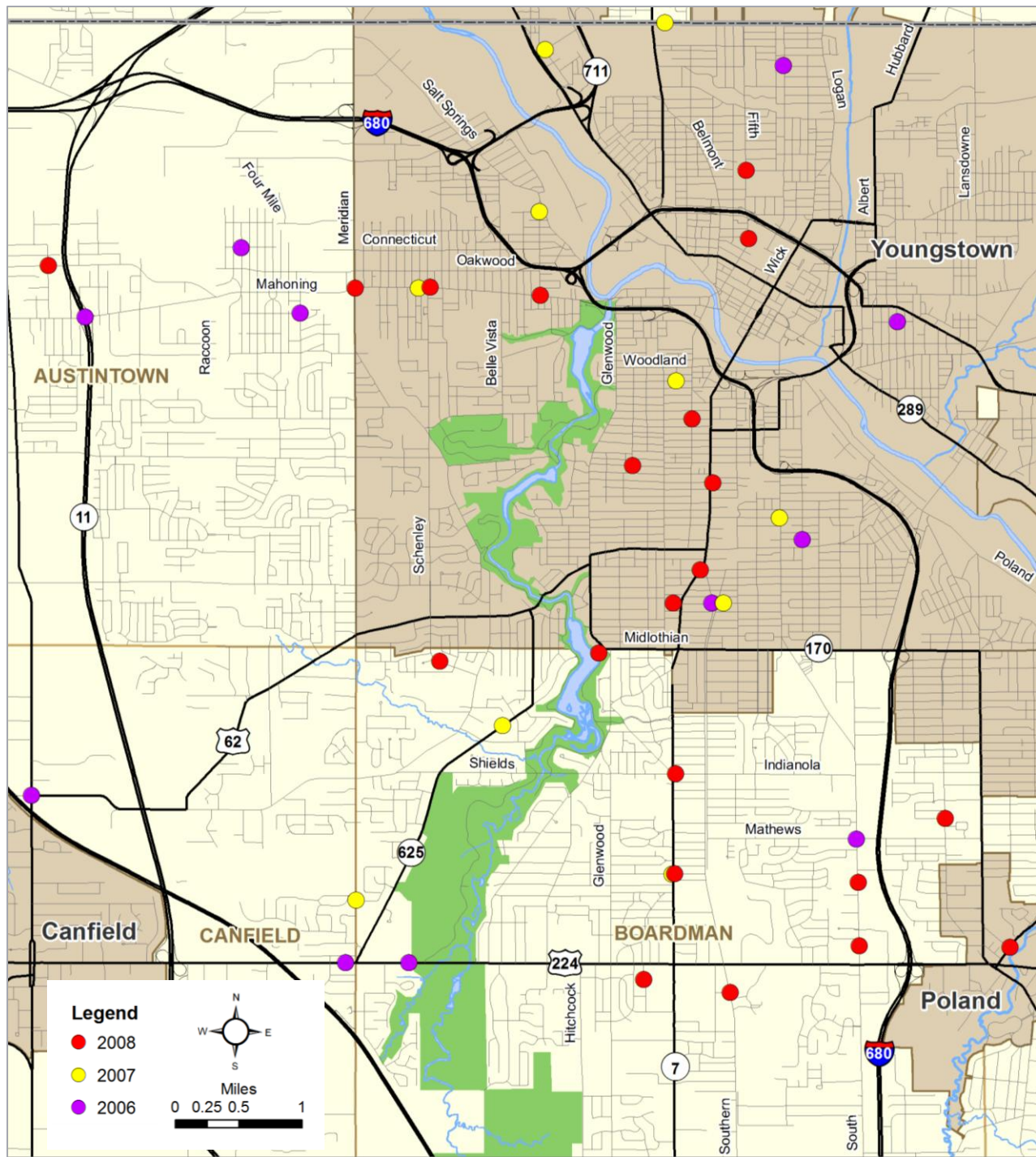
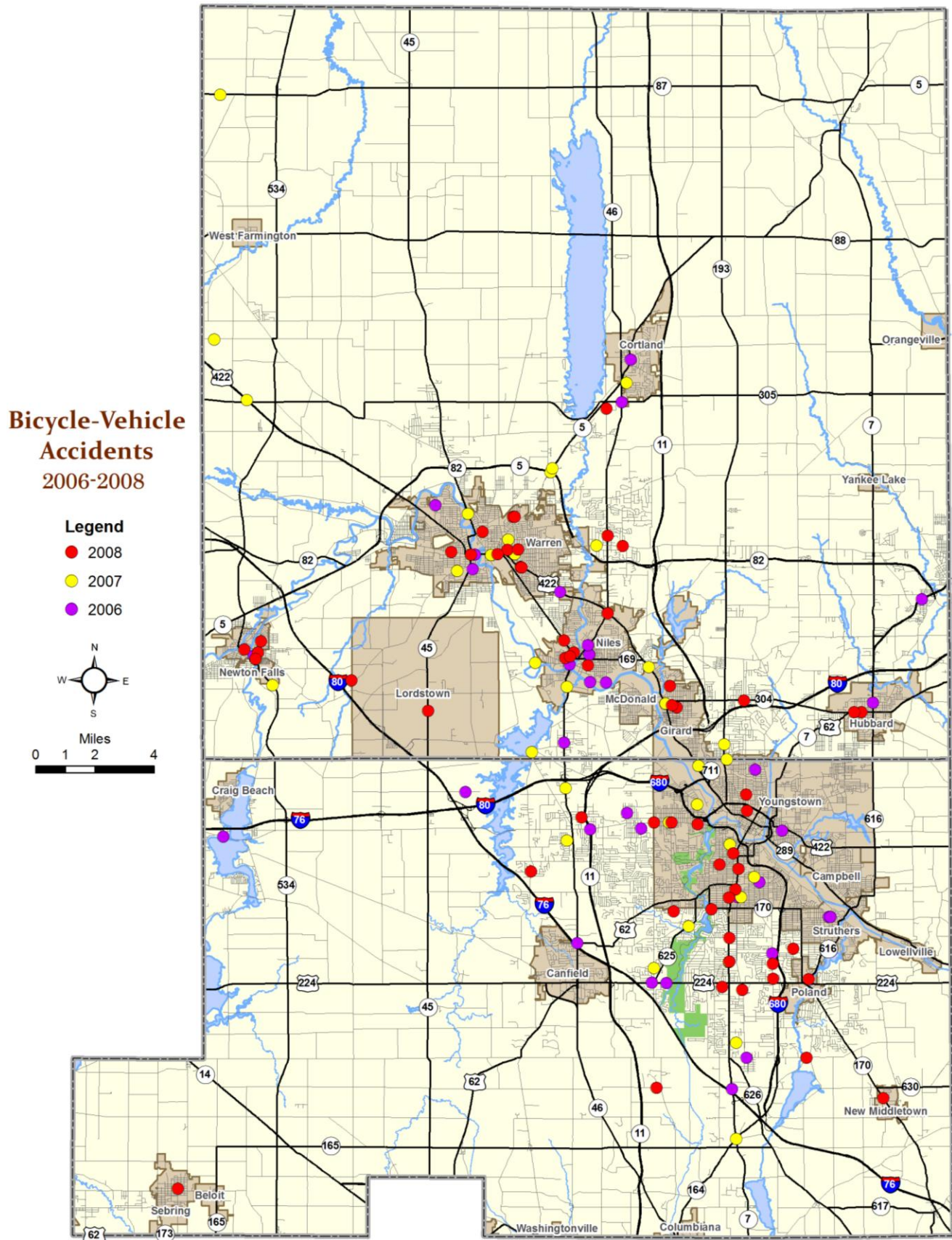


Figure 21: Bicycle-Vehicle Accidents, 2006-2008.



Planning and Funding

1|Introduction

As an MPO, Eastgate partners with local sponsors that apply for federal highway funding for a bicycle project. Eastgate does not sponsor or construct projects on its own. When an eligible project sponsor applies for federal highway funding from Eastgate for an eligible project, Eastgate will provide funding and help the sponsor ensure that planning and construction requirements stipulated by the federal government under Title 23 of the United States Code are met.

The Regional Bicycle Plan contains recommendations and a general plan for bicycle facility development in the area. The reality of property ownership, zoning, land use, location of environmental resources, availability of a local funding match, federal requirements, etc. may make some items recommended in the Regional Bicycle Plan ineligible or not feasible.

Eastgate will continue to be flexible when working with local communities in the development of bicycle facilities. Eastgate acknowledges that its member communities have their own priorities and needs and will support eligible bicycle projects brought to Eastgate, whether identified in the Regional Bicycle Plan or not.

Eastgate funds projects with federal highway dollars through several different programs. Each program has its own eligibility requirements, as set forth by the federal government. These programs, including the amount of funding available and the program's existence, are dependent on transportation bills passed by the federal government.

The last transportation bill was passed on August 10, 2005 when the President signed into law the Safe, Accountable, Flexible, Efficient Transportation Equity Act – A Legacy for Users (SAFETEA-LU). SAFETEA-LU was the largest surface transportation bill in our nation's history, providing \$286.5 billion in guaranteed funding for highways, highway safety and public transportation. SAFETEA-LU expired on September 30, 2009, but continues through resolution extensions that allow funding to continue at FY2009 levels while Congress decides on a new surface transportation bill.

“SAFETEA-LU expired on September 30, 2009, but continues through resolution extensions that allow funding to continue at FY2009 levels.”

2|Project Sponsors

To initiate a bicycle project, an eligible project sponsor must apply for federal funding from Eastgate. Eligible sponsors are acknowledged by ODOT as being capable of administering a project from preliminary development through construction. They have the qualified staff, experience, and management oversight necessary to meet all federal and state requirements.

Sponsors will be required to enter into a contract with ODOT for the implementation phase of the project. Any citizen group or other private organization may sponsor a project by coordinating with and making application through the eligible entity having jurisdiction over the transportation facility involved.

Sponsors must demonstrate that they have the financial resources to pay for cost overruns and the required local match, usually 20% of the total project cost. For example, a project estimated at a cost of \$1,000,000 may have \$800,000 (80%) paid by Eastgate. The remaining \$200,000 (20%) would be the responsibility of the project sponsor, in addition to any cost overruns. Sponsors must also maintain the transportation facility after construction.

Unlike some MPOs, Eastgate does not require the local match to come from a community's general fund. Local match may come from any non-federal funding source, with the exception of in-kind contributions. Every effort will be made to assist the project sponsor in finding a local match through other Eastgate or ODOT programs to minimize out-of-pocket costs.

3|The Federal-aid Highway System

All highways in the United States are functionally classified to provide a consistently defined roadway network across the country. The FHWA outlines criteria and ultimately approves classification, but it is ODOT, Eastgate, and local officials that initially determine the classification of our region's roadways.

The functional classification is tied to the Federal-aid Highway System and to eligibility for federal highway funding. Only roads on the Federal-aid Highway System can receive federal funding from Eastgate. Roads functionally classified as local streets or minor collectors are not part of the Federal-aid Highway System and are not normally eligible for federal highway funds. Roads eligible for federal funding are identified in Figure 22.

Figure 22: Federal-aid Highway System.



4|Federal Funding Programs

Eastgate receives about \$10 million in federal dollars each fiscal year to use for transportation projects. Eastgate typically funds bicycle projects through the Congestion Mitigation and Air Quality program. However it is also possible to fund a project through the Surface Transportation Program and Transportation Enhancement Activities. Each program has specific intents and eligibility requirements set by the federal government. Eastgate remains as flexible as possible in funding decisions, but must ensure that a project meets all applicable federal requirements to receive funding. Bicycle projects funded through Eastgate must serve a transportation purpose, rather than a recreational purpose.

Congestion Mitigation and Air Quality

The primary purpose of the Congestion Mitigation and Air Quality Improvement Program (CMAQ) is to fund projects and programs which reduce transportation related emissions.

Funding is available for:

- Transit and Public Transportation Programs
- Traffic Flow Improvements
- Travel Demand Management Strategies
- Ride Sharing Programs
- Pedestrian and Bicycle Programs
- Education and Outreach
- Inspection and Maintenance Programs
- Extreme Cold Start Programs
- Alternative "Clean" Fuels
- Public/Private Partnerships
- Experimental Pilot Projects
- Recipient Responsibilities
- Reporting Requirements

Figure 23: The MetroParks Bikeway. Image courtesy of Mill Creek MetroParks.



Projects funded through the Congestion Mitigation and Air Quality Program must be likely to contribute to the attainment of national ambient air quality standards. Bicycle projects are particularly suited for using CMAQ funding because no mobile source emissions are produced by travelers using bicycles. Projects may include bicycle facilities, bicycle parking, racks on buses, sidewalks, trails, and even promotional programs such as bike-to-work events. The Great Ohio Lake-to-River Greenway in Eastgate's area was constructed using CMAQ funding (Figure 23).

Surface Transportation Program

The Surface Transportation Program (STP) provides funds for construction, reconstruction, rehabilitation, resurfacing, restoration, and operational improvements for highways and bridges, including any such construction or reconstruction necessary to accommodate other transportation modes. Funds may be used for bicycle projects and include such items as on-road facilities, off-road paths, sidewalks, crosswalks, bicycle signals, parking or other ancillary facilities. Stand alone bicycle projects are eligible, but Eastgate and ODOT prefer bicycle projects to occur in conjunction with a programmed roadway project.

Transportation Enhancement

Transportation Enhancement (TE) funding is a 10% set aside of the STP. A TE project must relate to surface transportation by being in proximity to a highway or a pedestrian/bicycle corridor, enhancing the aesthetic, cultural or historic aspects of the travel experience, or serving a current or past transportation purpose.

The activities funded by TE includes:

- Provision of facilities for pedestrians and bicyclists
- Provision of safety and educational activities for pedestrians and bicyclists
- Acquisition of scenic easements and scenic or historic sites
- Scenic or historic highway programs
- Landscaping and other scenic programs
- Historic preservation

Common activities funded by the TE program include bicycle facilities and conversion of abandoned railroad corridors to bicycle trails.

Other Federal Sources

Transportation funding for bicycle projects is available through other federal sources besides Eastgate. Like Eastgate's programs, funding depends on the type, purpose and location of a bicycle project. Table 14 identifies which federal programs have funds available for bicycle-related projects.

Table 14: Federal funding opportunities (FHWA, 2008).

	NHS	STP	HSIP	SRTS	TEA	CMAQ	RTP	FTA	TE	BRI	402	PLA	TCSP	JOBS	FLH	BYW
Bicycle plan		•				•						•	•			
Bicycle lanes	•	•	•	•	•	•		•	•	•					•	•
Paved shoulders	•	•	•	•	•	•				•					•	•
Signed bike route	•	•		•	•	•									•	•
Shared use path	•	•		•	•	•	•			•					•	•
Single track trail							•									
Spot improvement program		•	•	•	•	•										
Maps		•		•		•					•					
Bike racks on bus		•			•	•		•	•							
Bicycle parking		•		•	•	•		•	•							•
Trail/highway intersection	•	•	•	•	•	•	•								•	•
Bicycle storage/service center		•		•	•	•		•	•				•	•		
Sidewalks	•	•	•	•	•	•		•	•	•					•	•
Crosswalks	•	•	•	•	•	•		•	•						•	•
Signal improvements	•	•	•	•	•	•										
Curb cuts/ ramps	•	•	•	•	•	•										
Traffic calming		•	•	•									•			
Coordinator position		•		•		•							•			
Safety/education position		•		•		•					•					
Police patrol		•		•							•					
Helmet promotion		•		•	•						•					
Safety brochure/book		•		•	•	•	•				•					
Training		•		•	•	•	•				•					

NHS	National Highway System	BRI	Bridge
STP	Surface Transportation Program	402	State and Community Traffic Safety Program
HSIP	Highway Safety Improvement Program	PLA	State/Metropolitan Planning Funds
SRTS	Safe Routes to School Program	TCSP	Transportation and Community and System Preservation Pilot Program
TEA	Transportation Enhancement Activities	JOBS	Access to Jobs/Reverse Commute Program
CMAQ	Congestion Mitigation/Air Quality Program	RTP	Recreational Trails Program
FLH	Federal Lands Highway Program	FTA	Federal Transit Capital, Urban & Rural Funds
BYW	Scenic Byways	TE	Transit Enhancements

5|Project Development Process

When a bicycle project is funded with federal money through Eastgate, certain requirements must be met and a certain process must be followed. A graphic depiction of ODOT's project development process for minor projects is shown in Figure 24. Depending on the scope of the project (major, minor, or minimal project), the process may be simpler or more complicated than the example. Eastgate will help the project sponsor work through the process, particularly the following steps:

MPO Long Range Plan Adoption

The proposed project will need to be added to Eastgate's Long Range Transportation Plan through a vote by Eastgate's governing bodies. The Long Range Transportation Plan is a fiscally constrained 20-year summary of expected transportation investments which includes both short-term and long-term projects that will support the development of an integrated intermodal transportation system.

Inclusion in the Transportation Improvement Program

The Transportation Improvement Program (TIP) is a 4-year detailed subset of the Long Range Plan. To receive federal funds, projects and programs must be in the financially constrained TIP. The project will be added to the TIP through a vote by Eastgate's governing bodies.

Inclusion into the State Transportation Improvement Program

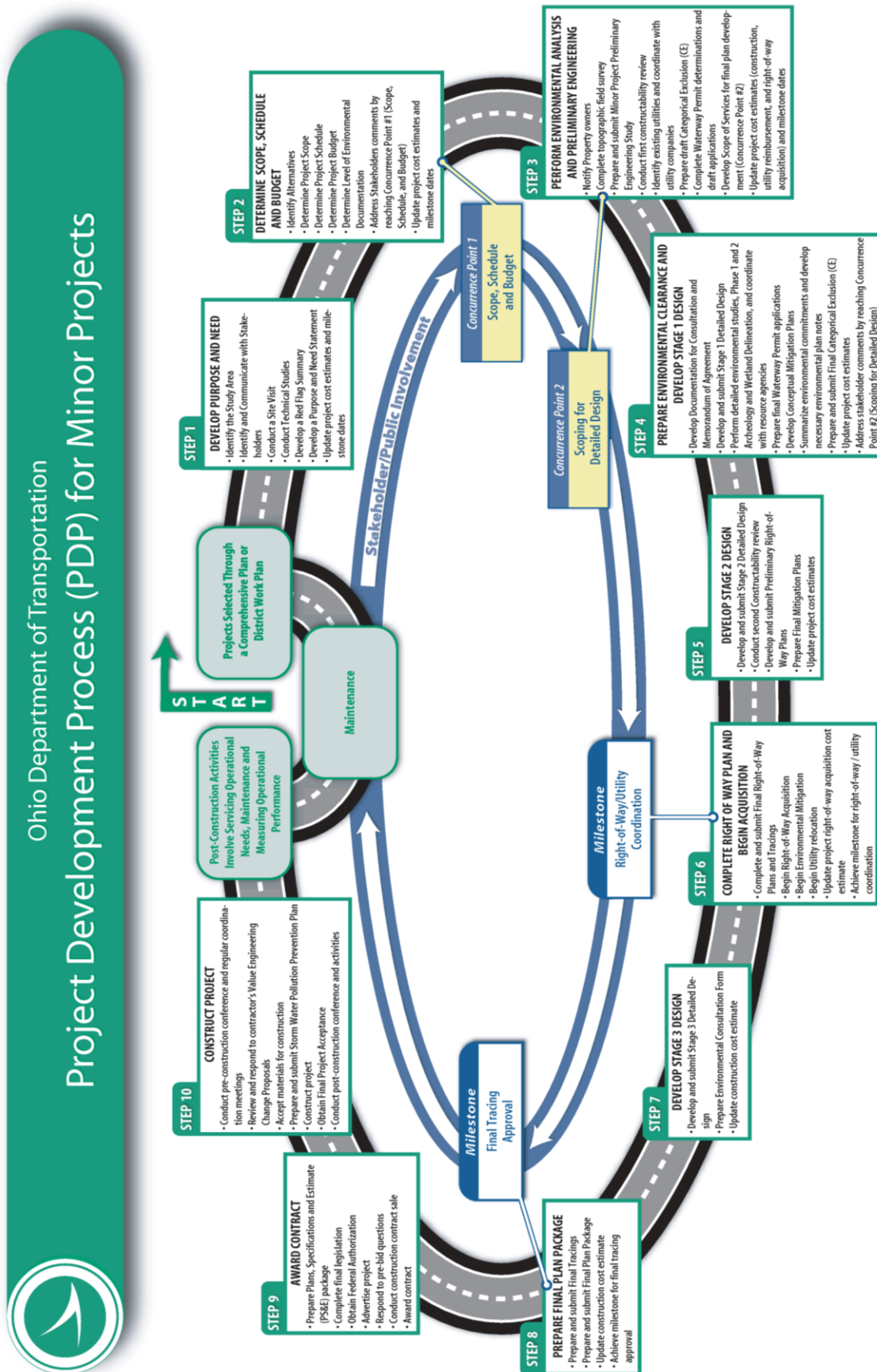
It is critical that the project also be included in the State Transportation Improvement Program (STIP). Existing procedures between Eastgate and ODOT automatically incorporate projects and programs from the TIP into the STIP, as long as they are consistent with the State Transportation Plan.

Local Support and Public Participation

Eastgate remains committed to a transportation planning process that includes a proactive public involvement process which provides complete information, timely public notice, full public access to key decisions, and supports early and continuing involvement of the public. Eastgate's Public Participation Program will recognize participants interested in or affected by proposed plans and programs and provide reasonable opportunities for interested parties to comment.

To insure implementation of the transportation plan it is important that local support be sustained after plan adoption. To ensure ongoing support local officials should continue to be involved, local jurisdictions should include the project in a locally adopted plan, and public involvement should be maintained.

Figure 24: ODOT Project Development Process (PDP) for Minor Projects.



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